

❖ PoWA (PostgreSQL Workload Analyzer).

❖ PoWA (PostgreSQL Workload Analyzer) is a tool that helps you see how your PostgreSQL database is performing.

1. It tracks slow or heavy queries.
2. It shows trends over time, not just current stats.
3. It helps find problems like high CPU, slow queries, or inefficient indexes.
4. It has a web dashboard for easy visualization.

In short: PoWA monitors your database and tells you what is slowing it down.

Step 1:- Prerequisites

● You need:

1. PostgreSQL $\geq$ 13 (works with 9.4+ but latest versions are recommended)
2. Superuser access
3. The following extensions installed:

- pwa
- pg_stat_statements
- pg_qualstats
- pg_stat_kcache
- pg_wait_sampling
- pg_track_settings
- Powa-web

Component	Purpose
pg_stat_statements	Tracks SQL execution statistics
pg_stat_kcache	OS-level CPU/IO stats
pg_qualstats	Tracks query filters for optimization
pg_wait_sampling	Monitors wait events
pg_track_settings	Tracks config changes
powa	Central aggregator + scheduler

To install this we used below site (according to postgresql version we have install extension)

1. Pg_stat_statements:-

Install through contrib package

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

postgresql16-contrib-16.0-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:54	727387
postgresql16-contrib-16.1-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:54	722969
postgresql16-contrib-16.1-2PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:54	732921
postgresql16-contrib-16.10-1PGDG.rhel9.x86_64.rpm	13-Aug-2025 12:36	747924
postgresql16-contrib-16.2-1PGDG.rhel9.x86_64.rpm	07-Feb-2024 20:49	733786
postgresql16-contrib-16.3-1PGDG.rhel9.x86_64.rpm	08-May-2024 12:46	741111
postgresql16-contrib-16.4-1PGDG.rhel9.x86_64.rpm	07-Aug-2024 11:47	741121
postgresql16-contrib-16.5-1PGDG.rhel9.x86_64.rpm	12-Nov-2024 21:56	741608
postgresql16-contrib-16.6-1PGDG.rhel9.x86_64.rpm	20-Nov-2024 18:20	741207
postgresql16-contrib-16.7-1PGDG.rhel9.x86_64.rpm	12-Feb-2025 12:17	748905
postgresql16-contrib-16.8-1PGDG.rhel9.x86_64.rpm	18-Feb-2025 12:51	747912
postgresql16-contrib-16.9-1PGDG.rhel9.x86_64.rpm	06-May-2025 10:05	748786
postgresql16-contrib-16.9-2PGDG.rhel9.x86_64.rpm	13-May-2025 08:00	749138
postgresql16-contrib-16.9-3PGDG.rhel9.x86_64.rpm	28-May-2025 08:00	748247

2. pg_stat_kcache:-

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

pg_stat_kcache-16-2.2.2-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	25843
pg_stat_kcache-16-2.2.3-1PGDG.rhel9.x86_64.rpm	29-Jan-2024 21:51	26220
pg_stat_kcache-16-2.3.0-1PGDG.rhel9.x86_64.rpm	17-Sep-2024 11:50	27041
pg_stat_kcache-16-2.3.1-1PGDG.rhel9.x86_64.rpm	27-Sep-2025 14:40	27137
pg_stat_kcache-16-llvmjit-2.2.2-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	30666
pg_stat_kcache-16-llvmjit-2.2.3-1PGDG.rhel9.x86_64.rpm	29-Jan-2024 21:51	30623
pg_stat_kcache-16-llvmjit-2.3.0-1PGDG.rhel9.x86_64.rpm	17-Sep-2024 11:50	27575
pg_stat_kcache-16-llvmjit-2.3.1-1PGDG.rhel9.x86_64.rpm	27-Sep-2025 14:40	27008

3. pg_qualstats:-

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

pg_profile-16-4.8-1PGDG.rhel9.noarch.rpm	17-Jan-2025 17:48	119827
pg_qualstats-16-2.0.4-3PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	35016
pg_qualstats-16-2.1.0-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	36191
pg_qualstats-16-2.1.1-1PGDG.rhel9.x86_64.rpm	26-Sep-2024 02:12	37007
pg_qualstats-16-llvmjit-2.0.4-3PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	42821
pg_qualstats-16-llvmjit-2.1.0-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	42897
pg_qualstats-16-llvmjit-2.1.1-1PGDG.rhel9.x86_64.rpm	26-Sep-2024 02:12	38458
pg_readme-16-0.7.0-1PGDG.rhel9.noarch.rpm	04-Sep-2024 07:28	31560
pg_readonly-16-1.0.3-2.rhel9.1.x86_64.rpm	02-Jan-2024 16:53	16943

4. pg_wait_sampling:-

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

pg_wait_sampling-16-1.1.5-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	22566
pg_wait_sampling-16-1.1.6-1PGDG.rhel9.x86_64.rpm	08-Aug-2024 11:21	24818
pg_wait_sampling-16-1.1.7-1PGDG.rhel9.x86_64.rpm	11-Feb-2025 08:41	24314
pg_wait_sampling-16-1.1.8-1PGDG.rhel9.x86_64.rpm	24-Feb-2025 09:48	24422
pg_wait_sampling-16-1.1.9-1PGDG.rhel9.x86_64.rpm	04-Aug-2025 23:05	24537
pg_wait_sampling-16-llvmjit-1.1.5-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	34477
pg_wait_sampling-16-llvmjit-1.1.6-1PGDG.rhel9.x86_64.rpm	08-Aug-2024 11:21	33724
pg_wait_sampling-16-llvmjit-1.1.7-1PGDG.rhel9.x86_64.rpm	11-Feb-2025 08:41	32706
pg_wait_sampling-16-llvmjit-1.1.8-1PGDG.rhel9.x86_64.rpm	24-Feb-2025 09:48	32784
pg_wait_sampling-16-llvmjit-1.1.9-1PGDG.rhel9.x86_64.rpm	04-Aug-2025 23:05	32389

5. pg_track_settings:-

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

pg_tle_16-llvmjit-1.5.0-1PGDG.rhel9.x86_64.rpm	22-Mar-2025 11:56	124105
pg_tle_16-llvmjit-1.5.1-1PGDG.rhel9.x86_64.rpm	09-May-2025 19:23	124257
pg_track_settings_16-2.1.2-2PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	16474
pg_uuidv7_16-1.2.0-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:53	20088

6. Powa:-

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

powa_16-4.2.0-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:54	6653
powa_16-4.2.1-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:54	6717
powa_16-4.2.2-1PGDG.rhel9.x86_64.rpm	02-Jan-2024 16:54	6797
powa_16-5.0.1-1PGDG.rhel9.x86_64.rpm	16-Dec-2024 09:46	6869

7. Powa-web:-

Website:- https://ftp.postgresql.org/pub/repos/yum/common/redhat/rhel-9.0-x86_64/

powa-web-4.2.0-1PGDG.rhel9.noarch.rpm	02-Jan-2024 16:22	621514
powa-web-5.0.1-1PGDG.rhel9.noarch.rpm	11-Dec-2024 12:18	560460
powa-web-5.0.2-1PGDG.rhel9.noarch.rpm	15-Jul-2025 14:13	567646
proj81-8.1.1-1.rhel9.x86_64.rpm	02-Jan-2024 16:23	2535892
proj81-devel-8.1.1-1.rhel9.x86_64.rpm	02-Jan-2024 16:23	94884

Differnce:-

❖ PGDG.rhel9.x86_64.rpm vs llvmjit-1.1.5-1PGDG.rhel9.x86_64.rpm:-

1. Normal version (example):-

- PGDG.rhel9.x86_64.rpm
- This is the normal version — it works on all PostgreSQL 16 servers.
- It collects information like:
 1. how much CPU your queries use
 2. how much memory or disk they touch
 3. It's simple and works everywhere.

✂ 2. "llvmjit" version (example)

- pg_stat_kcache_16-llvmjit-2.3.1-1PGDG.rhel9.x86_64.rpm
- This one is built for PostgreSQL servers that have JIT (Just-In-Time) turned on.

JIT is a feature that makes PostgreSQL run queries faster by compiling parts of them to machine code on the fly.

- So this package is for:

1. people using JIT
2. servers with postgresql16-llvmjit installed

If your PostgreSQL does not use JIT, you don't need this version.

Your PostgreSQL setup	Use this package
Normal PostgreSQL (no JIT)	pg_stat_kcache_16-2.3.1
PostgreSQL with JIT enabled	pg_stat_kcache_16-llvmjit-2.3.1

❖ Two different PostgreSQL repositories

🔗 1. common/pgdg-rhel9-extras:-

Website:- https://ftp.postgresql.org/pub/repos/yum/common/redhat/rhel-9.0-x86_64/

❖ Purpose: Extra tools and extensions

- This repo has common or extra packages that can work with any PostgreSQL version — like monitoring tools, helper utilities, or add-ons.

- Example:

1. Tools like powa-web, pgBadger, patroni, etc.
2. Not directly the PostgreSQL server itself.

❖ Use it when:

You want to install extra PostgreSQL-related tools, not the database server.

🔗 2. 16/redhat/rhel-9-x86_64/

Website:- https://ftp.postgresql.org/pub/repos/yum/16/redhat/rhel-9-x86_64/

❖ 🔗 Purpose: PostgreSQL 16 main packages

- This repo contains PostgreSQL 16 itself — the database server and its version-specific extensions.

❖ Example:

postgresql16-server

postgresql16-contrib

pg_cron_16, pg_stat_statements_16, etc.

❖ **Use it when:**

You are installing or upgrading to PostgreSQL 16.

✓ **In short:**

Repo	What it's for	Example packages
pgdg-rhel9-extras	Extra tools/extensions (works with many versions)	powa-web, patroni
16/redhat/rhel-9-x86_64	PostgreSQL 16 main server & extensions	postgresql16-server, pg_cron_16

Step 2:- Install Extension , create database and create extension:-

```
root@localhost:~/extension
[root@localhost extension]# ll
total 1408
-rw-r--r--. 1 root root 37007 Sep 26 2024 pg_qualstats_16-2.1.1-1PGDG.rhel9.x86_64.rpm
-rw-r--r--. 1 root root 27137 Sep 27 20:10 pg_stat_kcache_16-2.3.1-1PGDG.rhel9.x86_64.rpm
-rw-r--r--. 1 root root 16474 Jan 2 2024 pg_track_settings_16-2.1.2-2PGDG.rhel9.x86_64.rpm
-rw-r--r--. 1 root root 24537 Aug 5 04:35 pg_wait_sampling_16-1.1.9-1PGDG.rhel9.x86_64.rpm
-rw-r--r--. 1 root root 748247 May 28 13:30 postgresql16-contrib-16.9-3PGDG.rhel9.x86_64.rpm
-rw-r--r--. 1 root root 6869 Dec 16 2024 powa_16-5.0.1-1PGDG.rhel9.x86_64.rpm
-rw-r--r--. 1 root root 567646 Jul 15 19:43 powa-web-5.0.2-1PGDG.rhel9.noarch.rpm
[root@localhost extension]# dnf install * -y
```

Error:-

```
[root@localhost ~]# powa-web
Traceback (most recent call last):
  File "/usr/bin/powa-web", line 6, in <module>
    from powa import make_app
  File "/usr/lib/python3.9/site-packages/powa/_init__.py", line 24, in <module>
    from powa.collector import (
  File "/usr/lib/python3.9/site-packages/powa/collector.py", line 9, in <module>
    from powa.dashboards import MetricGroupDef
  File "/usr/lib/python3.9/site-packages/powa/dashboards.py", line 8, in <module>
    from powa.compat import classproperty, with_metaclass
  File "/usr/lib/python3.9/site-packages/powa/compat.py", line 12, in <module>
    import pycopg2
ModuleNotFoundError: No module named 'pycopg2'
[root@localhost ~]#
```

Sol:-

Your system is trying to run powa-web, but Python cannot find the psycopg2 module. That's why it crashes:

ModuleNotFoundError: No module named 'psycopg2'

psycopg2 is the Python PostgreSQL driver — it lets PoWA connect to your PostgreSQL server.

❖ How to fix it

1🔗 Install psycopg2 for your Python version (Python 3.9 in your case)

On RHEL 9 / CentOS 9, you can do:

```
sudo dnf install python3-psycopg2
```

Or, using pip:

```
sudo python3 -m pip install psycopg2-binary
```

psycopg2-binary is usually easier because it comes precompiled.

2🔗 Verify installation

```
python3 -c "import psycopg2; print(psycopg2.__version__)"
```

3. Enable required extensions

Edit your PostgreSQL configuration file (postgresql.conf):

```
shared_preload_libraries = 'pg_stat_statements, pg_stat_kcache, pg_qualstats, pg_wait_sampling, powa'
```

4.If you don't have a dedicated powa database yet:

```
CREATE DATABASE powa;
```

5.Create extensions in your database

```
CREATE EXTENSION pg_stat_statements;
```

```
CREATE EXTENSION pg_qualstats;
```

```
CREATE EXTENSION pg_stat_kcache;
```

```
CREATE EXTENSION pg_wait_sampling;
```

```
CREATE EXTENSION hypopg; -----(PoWA needs the hypopg extension in all databases  
of the cluster in order to check that the suggested  
indexes are efficient:)
```

```
powa=# CREATE EXTENSION powa;  
ERROR: required extension "btree_gist" is not installed
```

HINT: Use CREATE EXTENSION ... CASCADE to install required extensions too.

CREATE EXTENSION powa CASCADE; ----- The keyword CASCADE tells PostgreSQL:
"Also install any other extensions that PoWA needs."

```
powa=# \dx

```

Name	Version	Schema	List of installed extensions	Description
btree_gist	1.7	public	support for indexing common datatypes in GiST	
hypopg	1.4.1	public	Hypothetical indexes for PostgreSQL	
pg_qualstats	2.1.1	public	An extension collecting statistics about qual	
pg_stat_kcache	2.3.1	public	Kernel statistics gathering	
pg_stat_statements	1.10	public	track planning and execution statistics of al	
pg_wait_sampling	1.1	public	sampling based statistics of wait events	
plpgsql	1.0	pg_catalog	PL/pgSQL procedural language	
powa	5.0.3	public	PostgreSQL Workload Analyser-core	

```
(8 rows)

powa=#
```

Step 3:- create dedicated user:-

- ❖ One last step is to create a role that has superuser privileges and is able to login to the cluster (use your own credentials):

CREATE ROLE powa SUPERUSER LOGIN PASSWORD 'astrongpassword' ;

Step 4- Configure the PoWA web interface

PoWA has a web dashboard called powa-web.

Edit its config file:

```
/etc/powa-web.conf

# Web server settings
listen_address = "0.0.0.0"
listen_port = 8888
debug = False
gzip = True
compress_response = True

# Cookie secret for secure sessions (generate a random string)
cookie_secret = "CHANGE_THIS_TO_A_RANDOM_SECRET"

# Servers to monitor
servers = {
    'main': {
        'host': '192.168.31.128', # PostgreSQL host
        'port': '5432',          # PostgreSQL port
        'database': 'powa',      # PoWA repository database
        'username': 'powa',      # DB user
        'password': 'astrongpassword' # DB password
    }
}
```

```
}  
}
```

❖ Then start the web UI:

```
sudo systemctl start powa-web  
sudo systemctl enable powa-web
```

Access it at:

🔗 <http://<your-server-ip>:8888>

Step 5:- Check PoWA status

Inside psql:

```
SELECT * FROM powa_servers;
```

```
postgres=# \c powa  
You are now connected to database "powa" as user "postgres".  
powa=# SELECT * FROM powa_servers;  
 id | hostname | alias | port | username | password | dbname | frequency | powa_coalesce | retention | allow_ui_connection | version  
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----  
  0 |         |      |    0 |         |         |      |      -1 |         100 | 00:00:00 | t                  |  
(1 row)  
  
powa=#
```

❖ You should see the monitored server listed.

You can also manually add a server:

```
SELECT powa_register_server(  
    hostname => '127.0.0.1',  
    port => 5432,  
    username => 'postgres',  
    password => 'your_password'  
);
```

Step 6:- Start collecting stats

❖ By default, PoWA collects data every 5 minutes.

❖ You can tune it:

```
ALTER SYSTEM SET powa.frequency = '1min';  
SELECT pg_reload_conf();
```

✔ Verification

Run:

```
SELECT * FROM powa_snapshot;
```

